UNIVERSITY OF SINDH

JAMSHORO, SINDH—PAKISTAN

B.Ed. (04 Years)

Course No: GQUR-301 Credit Hours **3**

Title of Course: **Quantitative Reasoning–II** Semester 2nd

Course Description

Quantitative Reasoning (II) is an introductory level undergraduate course that focuses on the fundamentals related to quantitative concepts and analysis. The course is designed to familiarize students with the basic concepts of statistics and to develop students' abilities to analyze and interpret quantitative information. Through a combination of theoretical concepts and practical exercise, this course will also enable students to cultivate their quantitative literacy and problem-solving skills while effectively expanding their academic horizon and breadth of knowledge of their specific major/field of study.

Course Outcome

By the end of this course, students shall have:

1. Understanding of fundamental statistical concepts.
2. Basic ability to interpret data presented in various formats including tables, graphs and charts.
3. It is an ability to critically evaluate quantitative information to make evidence-based decisions through appropriate computational tools.

Course Outline

Chapter 1: Introduction to Data Week 1

1.1 Types of Data
   a. Qualitative vs Quantitative
   b. Discrete vs Continuous

1.2 Sources of Data
   a. Primary Sources
   b. Secondary Sources

Chapter 2: Data Organization and Presentation Week 2

2.1 Tabular Presentation of Data
   a. Frequency Distribution Tables
   b. Cumulative Frequency

2.2 Graphical Representation
   a. Bar Charts
   b. Histograms
   c. Pie Charts

Chapter 3: Population and Sampling Week 3

3.1 Population Concepts
3.2 Sample and Sampling Techniques
   a. Random Sampling
   b. Non-Random Sampling

Chapter 4: Measures of Central Tendency – I Week 4

4.1 Mean (Arithmetic Average)
4.2 Median

Chapter 5: Measures of Central Tendency – II Week 5

5.1 Mode
5.2 Comparison of Mean, Median, Mode
5.3 Applications and Interpretation

Chapter 6: Measures of Dispersion Week 6

6.1 Range
6.2 Variance and Standard Deviation
6.3 Data Interpretation using Dispersion

Chapter 7: Counting Techniques Week 7

7.1 Basic Counting Principles
7.2 Multiplicative Rule

Chapter 8: Permutations and Combinations Week 8

8.1 Permutations
8.2 Combinations
8.3 Applications in Problem Solving

Chapter 9: Introduction to Probability Week 9

9.1 Basic Probability Concepts
9.2 Probability Rules

Chapter 10: Random Variables and Distributions Week 10

10.1 Random Variables (Discrete & Continuous)
10.2 Probability Distributions

Chapter 11: Probabilistic Models Week 11

11.1 Introduction to Probabilistic Models
11.2 Applications in Real Life

Chapter 12: Bivariate Analysis Week 12

12.1 Scatter Plots
12.2 Interpretation of Relationships

Chapter 13: Correlation and Regression Week 13
13.1 Correlation (Types & Coefficient)
13.2 Regression Analysis (Linear Regression)

Chapter 14: Estimation and Confidence Interval Week 14
14.1 Point Estimation
14.2 Confidence Intervals

Chapter 15: Hypothesis Testing Week 15
15.1 Concept of Hypothesis Testing
15.2 z-test
15.3 t-test

Chapter 16: Quantitative Reasoning Applications Week 16
16.1 Statistical Modeling Exercises
16.2 Problem Solving using Statistical Concepts
16.3 Review and Integration of Concepts

Teaching Strategy

The course will be delivered through a combination of lectures, class discussions, question–answer sessions, and brainstorming activities to promote active learning. Student-teachers will also engage in practical work, including group work, group assignments, individual assignments, and presentations, to enhance understanding and develop collaborative and analytical skills.

ASSESSMENT CRITERIA

Mid-Term marks	Final Term	Assignment/ Presentation	Attendance	Total
30	50	10	10	100

SUGGESTED INSTRUCTIONAL/READING MATERIALS

1. "Introductory Statistics" by Prem S. Mann.
2. "Applied Statistical Modeling" by Salvatore Baboons.
3. "Quantitative Reasoning: Thinking in Numbers" by Eric Zaslow.
4. "Statistics for Technology: A course in Applied Statistics" by Chatfield, C.
5. "Statistics: Unlocking the Power of Data" by Robin H. Lock, Patti Frazer Lock, Kari Lock Morgan, and Eric F. Lock
6. Chaudhry, S.M.and Kamal, S. (2008), "Introduction to Statistical Theory" Parts I &

II, 8th ed, Ilmi Kitab Khana, Lahore, Pakistan.
7. Haq, M. (1984). Foundation of Probability and Statistics, Tahir sons, Urdu Bazar, Karachi.